higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

ENGINEERING GRAPHICS AND DESIGN
(Second Paper)
NQF LEVEL 3

(6021023)

26 November 2019 (X-Paper)
09:00–13:00

REQUIREMENTS: A2 drawing sheet

Drawing instruments may be used.

This question paper consists of 4 pages and 5 diagram sheets.

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TIME: 4 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Use both sides of the DRAWING SHEET.
5. Draw a 20 mm border on both sides of the DRAWING SHEET.
6. ALL drawings, including the candidate's information, must be done in pencil.
7. A balanced layout is important and candidates will be penalised for poor planning.
8. A radius curve stencil may be used to draw arcs.
9. Use your own discretion where dimensions are not given.
10. ALL drawing work must conform to the latest SABS 0111 Code of Practice for Engineering Drawing.
11. Write neatly and legibly.

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QUESTION 1

DIAGRAM SHEET 1 (attached) shows a table.

1.1 Draw the table, shown in DIAGRAM SHEET 1 and print the answers to the following questions, using 3 mm lettering. (2)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and print only 'True' or 'False' next to the question number (1.2.1–1.2.4) in the table.

1.2.1 A three-dimensional view is a pictorial view, where two sides are visible.

1.2.2 An oblique axis is a line which is at 45 degrees, either to the left or right of the horizontal axis.

1.2.3 A vertical axis is a line or a centre line positioned on the vertical plane.

1.2.4 An isometric scale is drawn using a 45-degree set-square.

(4 × 2) (8)

[10]

QUESTION 2

DIAGRAM SHEET 2 (attached) shows TWO views of a casting in first-angle orthographic projection.

Do NOT copy the given views.

2.1 Construct an isometric scale. (2)

2.2 Draw an isometric view of the casting using the isometric scale, in first-angle orthographic projection. 'A' must be the lowest point of the isometric drawing. NO hidden detail required. (13)

[15]

QUESTION 3

DIAGRAM SHEET 3 (attached) shows a truncated, hexagonal pipe, with both ends open.

3.1 Draw the given views to scale 1:1. (4)

3.2 Develop the surface of the hexagonal pipe. (13)

[17]

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QUESTION 4

DIAGRAM SHEET 4 (attached) shows TWO views of a stuffing box on a pump, in third-angle orthographic projection.

Draw detailed drawings of the following items, to scale 1:2.

4.1 A full-sectional front view of item 1 (16)

4.2 A full-sectional front view of item 2 (8)

[24]

QUESTION 5

DIAGRAM SHEET 5 (attached) shows the components of Cable Pulley Unit in third-angle orthographic projection. The parts list is as follows:

ITEM	NAME	NO PER ASSY
1	Slide plates	2
2	Pulley	1
3	Special bolt	1
4	Eyepiece	1
5	M6 hexagonal nut	1 (Not shown)
6	M4 × 30 mm hexagonal bolt and nut	2 (Not shown)

5.1 Draw a full-sectional front view of an assembly drawing, to scale 2:1. (27)

5.2 Insert the item numbers on the assembly drawing. (5)

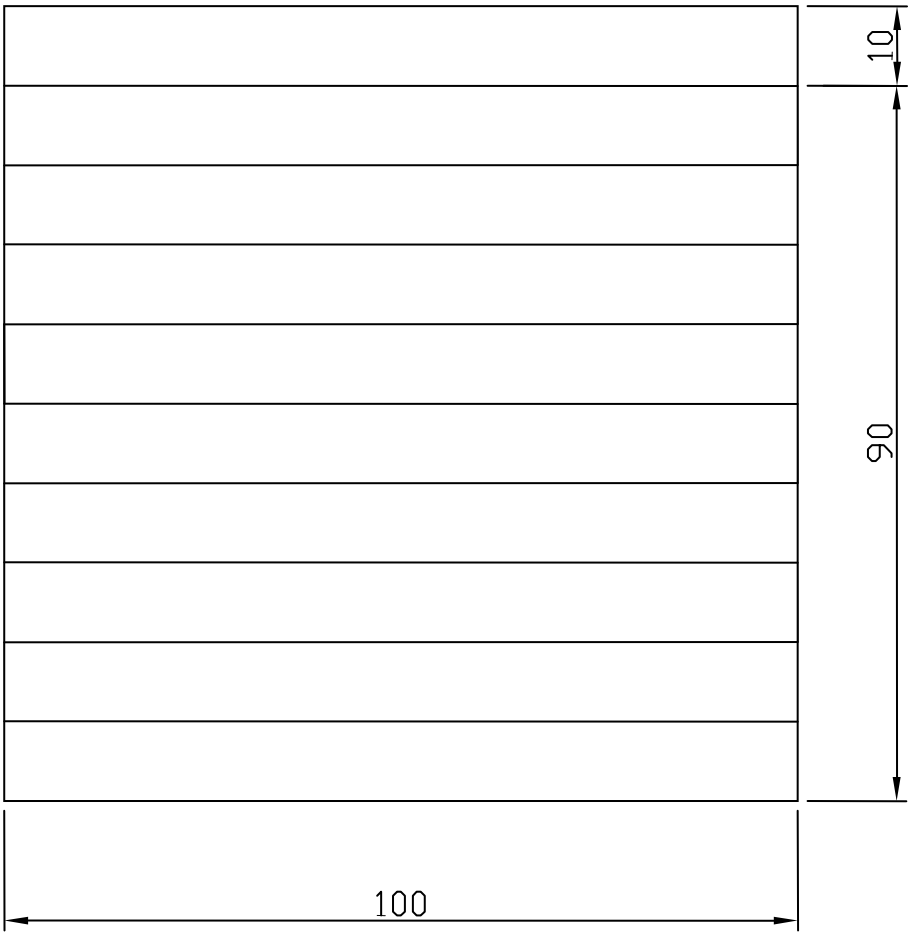
5.3 Insert the title and scale, centrally, below the assembly drawing. (2)

[34]

TOTAL: 100

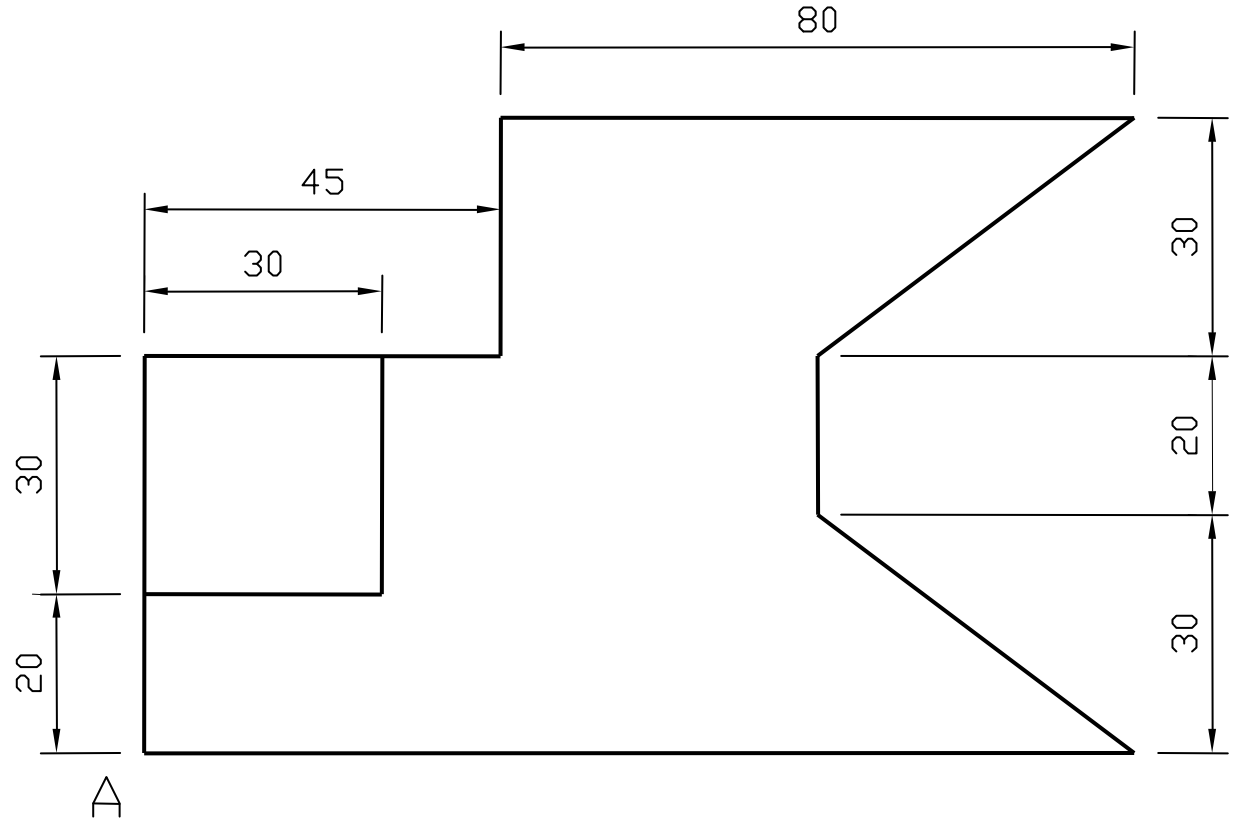
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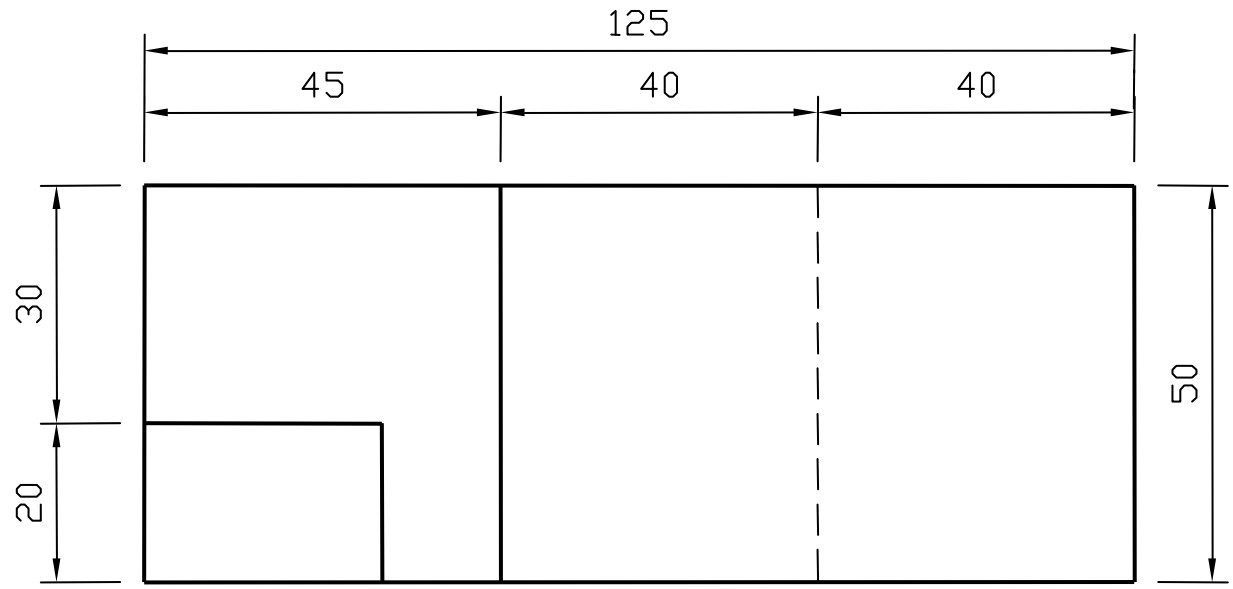
DIAGRAM SHEET 1



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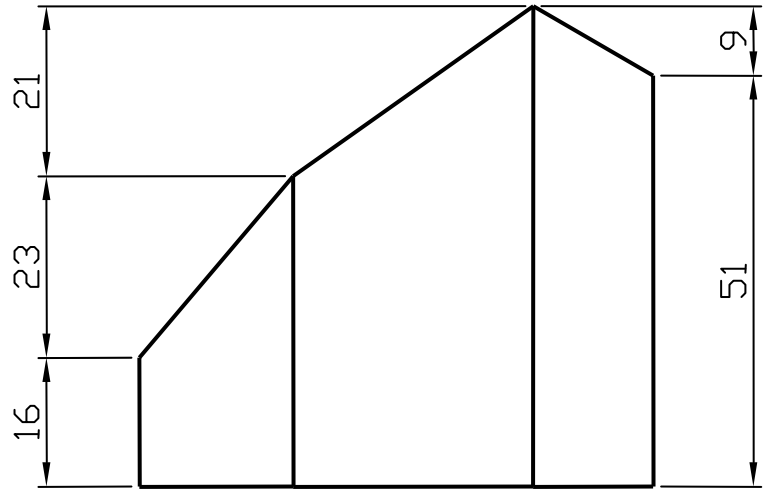
DIAGRAM SHEET 2

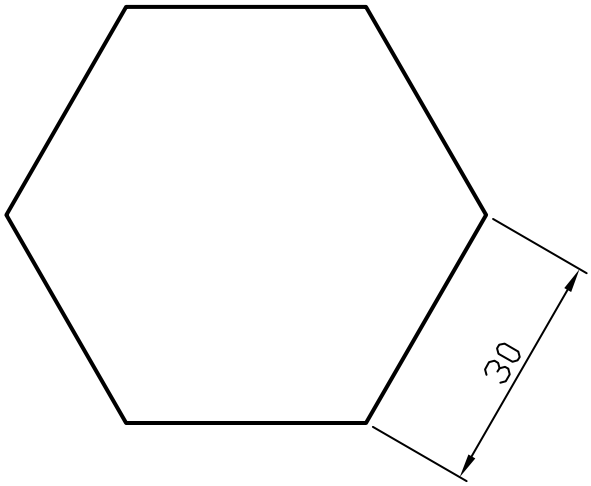




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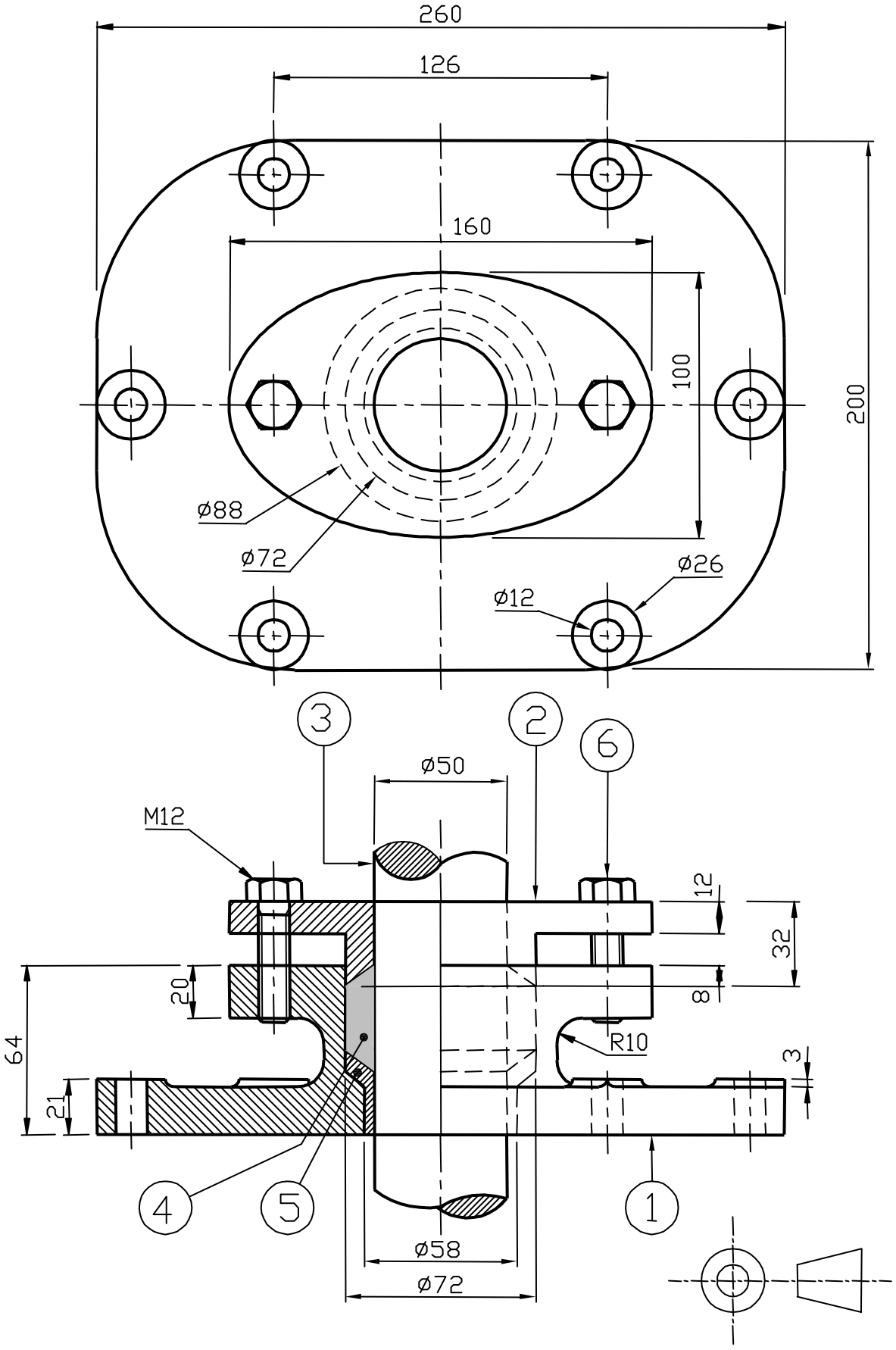
DIAGRAM SHEET 3





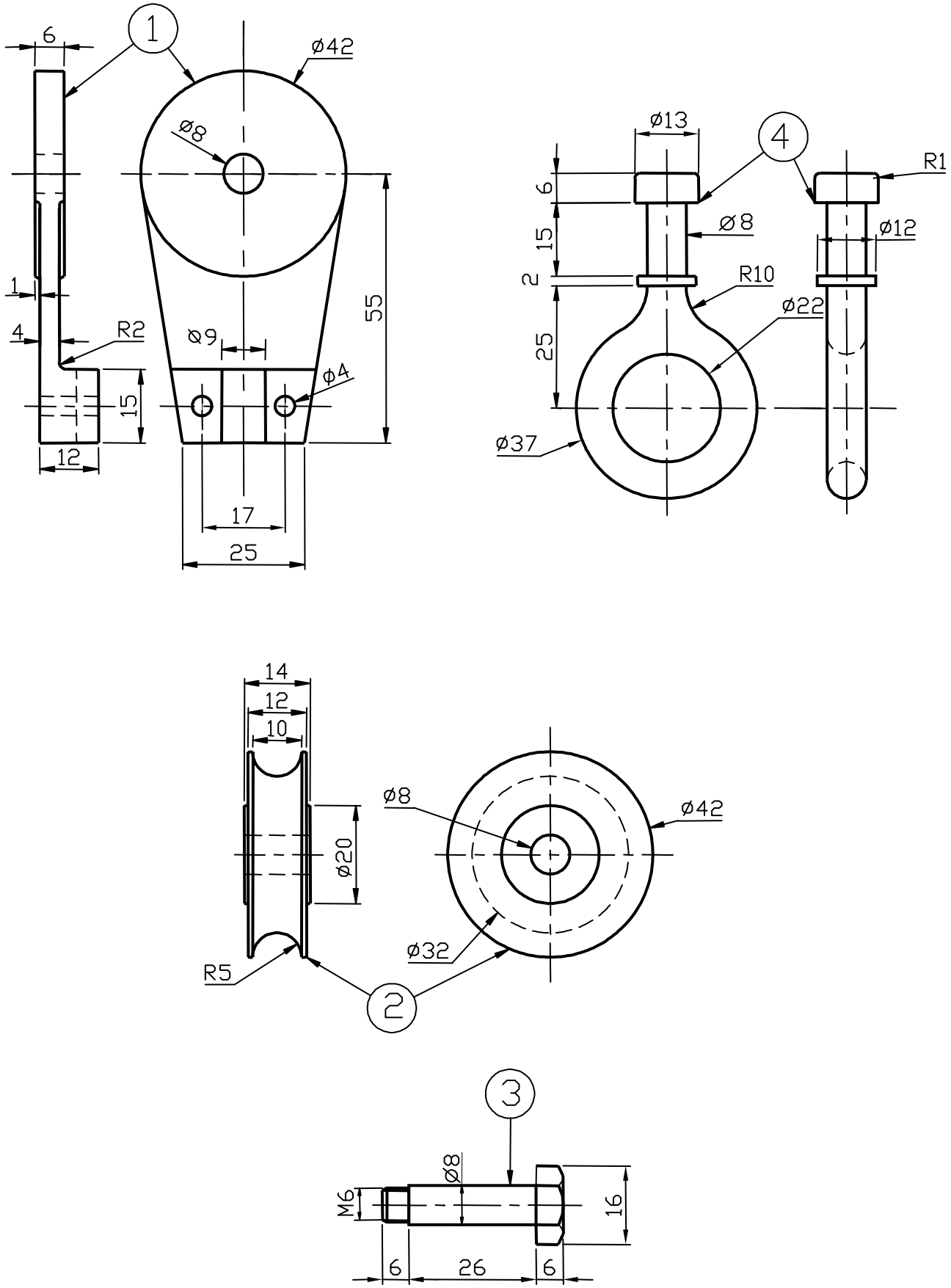
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DIAGRAM SHEET 4



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DIAGRAM SHEET 5



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